CBSE Class 10 Mathematics

(Standard) Previous Year Question

Paper – Set 2025

Time Allowed: 30 MiNUTES

General Instructions

- All questions are compulsory.
- Internal choice is provided wherever applicable.
- Questions follow the current CBSE syllabus.

SECTION A – Objective Questions

1. If one zero of the quadratic polynomial $f(x) = x^2 + 5x + k$ is 2, then the value of k is:

(a) −14

(b) 4

(c) −10

(d) 10

2. The pair of equations $2x + 3y = 7$ and $4x + 6y = 14$ has:-

(a) Infinitely many solutions.

(b) A unique solution.

(c) No solution.

(d) Two distinct solutions.

3. In an AP, the common difference is 2 and the first term is 3. The fifth term is:-

(a) 9

(b) 11

(c) 13

(d) 7

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4. Which of the following numbers is irrational?

(a) 0.36

(b) 2/7

(c) $\sqrt{7}$

(d) 3/5

5. The discriminant of the equation $x^2 - 4x + 4 = 0$ is:-

(a) 16

(b) 12

(c) 0

(d) 4

6. If $\sin\theta = 3/5$, then $\cos\theta$ is:-

(a) 4/5

(b) 3/4

(c) 5/3

(d) 1/5

7. The sum of the first 15 natural numbers is:-

(a) 120

(b) 105

(c) 115

(d) 140

8. If tangents from an external point A to a circle with center O touch the circle at P and Q, then OA is equal to:-

(a) AP

(b) AQ

(c) OP

(d) OQ

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9. The product of two consecutive positive integers is 132. The integers are:-

(a) 11, 12

(b) 10, 13

(c) 12, 13

(d) 6, 22

10. Which of the following is not a quadratic equation?

(a) $2x^2 - 4x + 3 = 0$

(b) $x + 5 = 0$

(c) $x^2 + 6x - 9 = 0$

(d) $3x^2 + 2x = 0$

11. The HCF of 20 and 28 is:-

(a) 4

(b) 7

(c) 2

(d) 5

12. Area of a circle with radius r is:-

(a) πr

(b) πr^2

(c) $2\pi r$

(d) $\pi/2$

13. On throwing a dice, the probability of getting an even number is:-

(a) 1/2

(b) 1/3

(c) 1/6

(d) 2/3

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14. Which of the following is the solution of the equation $2x - 3 = 9$?

(a) $x = 3$

(b) $x = 6$

(c) $x = -6$

(d) $x = -3$

15. If the ratio of the areas of two circles is 9:16, then the ratio of their radii is:-

(a) 3:4

(b) 9:16

(c) 4:3

(d) 16:9

16. Given that two lines with equations $3x - 2y = 7$ and $6x - 4y = 14$ are:-

(a) Intersecting

(b) Coincident

(c) Parallel

(d) Perpendicular

17. The next term in the sequence 8, 15, 22, ... is:-

(a) 27

(b) 29

(c) 33

(d) 38

18. If the perimeter of a square is 40 cm, its area is:

(a) 100 cm^2

(b) 36 cm^2

(c) 64 cm^2

(d) 25 cm^2

19. Assertion (A): The sum of four odd numbers is even.

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Reason (R): Even + Even = Even.

(a) Both A and R are true, and R is the correct explanation of A

(b) Both A and R are true, but R is not the correct explanation of A

(c) A is true, R is false

(d) A is false, R is true

20. If tan θ = 1, then θ is equal to:-

(a) 30°

(b) 45°

(c) 60°

(d) 90°

BHAYAT